## Learning Objectives

At the end of this lesson you shall be able to

- Understand fluids and its properties.
- List the types of pressure.
- Know the pressure measuring devices.
- Explain the different types of pipe flow.
- Understand Orifice, mouthpiece and its types.
- List the Classification of pumps.
- Explain the working of pumps

# 7.1 Introduction

## 7.1.1 Hydraulics – Definition

The branch of science, which deals with the properties and behavior of liquids is called 'Hydraulics'. It is otherwise called 'Hydromechanics'.

## 7.1.2 Fluids

Fluids are substances which are capable of flowing. They confirm to the shape of the container. They change their shape even under the action of very small forces. They have some degree of compressibility and offer little resistance to change of form. They include liquids and gases.

## 7.1.3 Properties of Fluids

The following are the some of the properties of fluids.

1. Density
2. Specific weight
3. Specific gravity
4. Cohesion
5. Adhesion
6. Surface tension
7. Capillarity
8. Viscosity

### 1. Density

Density is defined as the mass per unit volume. It is also termed as mass density or specific mass. It is usually denoted by the Greek letter $\rho$ (rho).

$$\text{Density} = \frac{mass}{volume} \quad \text{i.e., } \rho = \frac{m}{v}$$

Unit: $kg/m^3$

Density of pure water is 1000 $kg/m^3$.

### 2. Specific weight

Specific weight is defined as the weight per unit volume. It is otherwise known as weight density. It is denoted by the Greek letter $\gamma$ (gamma). Specific weight changes from place to place.

$$\text{Specific weight} = \frac{\text{Weight}}{\text{Volume}} \quad \text{i.e., } \gamma = \frac{w}{v}$$

Unit: $N/m^3$

Specific weight of pure water is 9.81 $kN/m^3$.

### 3. Specific Gravity

Specific gravity is defined as the ratio of mass density of liquid to the



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